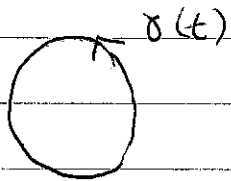


Return to finish up limit cycles
& their linear stability

1. Linearize eqns. about limit cycle $\gamma(t)$



$$\dot{x} = \cancel{f}(x)$$

$$x \in \mathbb{R}^n, n \geq 2$$

$$\dot{\gamma} = v(\gamma(t)), \quad \gamma(t) = \gamma(t+T)$$

Linearized eqn. $x(t) = \gamma(t) + y(t)$

$$\Rightarrow \dot{y} = A(t)y, \quad \text{where } A(t) = A(t+T),$$

since $A(t) = D_x v(\gamma(t))$

Fundamental matrix $\Phi(t)$ solves

$$\frac{d\Phi}{dt} = A(t)\Phi, \quad \Phi(0) = \text{Id.}$$

then if $y(0) = y_0$, $y(t) = \Phi(t)y_0$

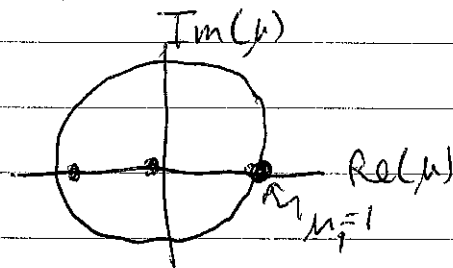
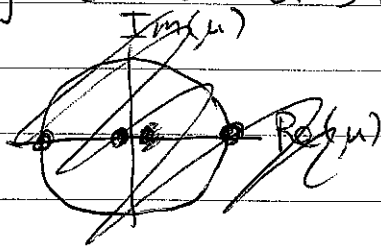
$$y(T) = \Phi(T)y_0, \quad y(2T) = \Phi(T)y(T) = M^2 y_0, \dots$$

$$= M y_0$$

↑ Monodromy matrix $M = \Phi(T)$

eigenvalues of M determine linear stability properties of $\gamma(t)$. There is necessarily 1 eigenvalue of M , $\mu = 1$, which is associated with perturbation along limit cycle.

If eigenvalues (= Floquet multipliers) are $\mu_1 = 1, \mu_2, \mu_3, \dots, \mu_n$ and $|\mu_j| < 1$ $j=2, \dots, n$, then limit cycle is asymptotically stable. If any μ_j has $|\mu_j| > 1$, then limit cycle is unstable.

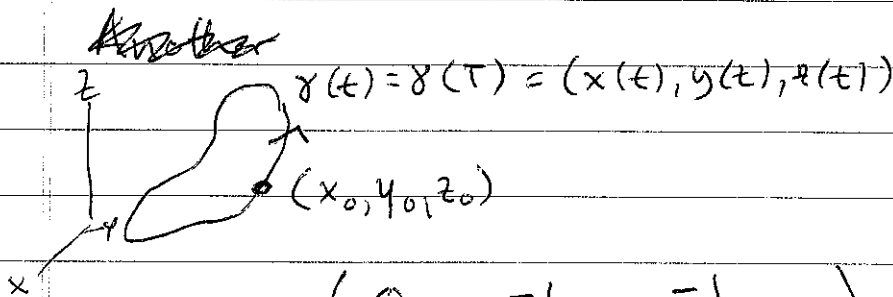


Rössler limit cycle

$$(*) \begin{cases} \dot{x} = -y - z \\ \dot{y} = x + ay \\ \dot{z} = b + z(x - c) \end{cases}$$

$$a = b = 0.2$$

$$c = 2.5 \nearrow$$



$$A(t) = \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z(t) & 0 & x(t) - c \end{pmatrix} \quad \text{for}$$

Perturbation $\begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix}$ ie. $\begin{pmatrix} \delta \dot{x} \\ \delta \dot{y} \\ \delta \dot{z} \end{pmatrix} = A(t) \begin{pmatrix} \delta x \\ \delta y \\ \delta z \end{pmatrix} \quad (*)$

To get M , simultaneously solve $(*)$ & $(**)$ with initial conditions $(x_0, y_0, z_0, 1, 0, 0)$ for $(x_0, y_0, z_0, 0, 1, 0)$ $(x_0, y_0, z_0, 0, 0, 1)$

time T (numerically)

$$\Phi_T \begin{pmatrix} x_0 \\ y_0 \\ z_0 \\ x \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_0 \\ y_0 \\ z_0 \\ a \\ b \\ c \end{pmatrix}, \quad \Phi_T \begin{pmatrix} x_0 \\ y_0 \\ z_0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_0 \\ y_0 \\ z_0 \\ d \\ e \\ f \end{pmatrix} \text{ etc}$$

$$M = \begin{pmatrix} a & d & g \\ b & e & h \\ c & f & i \end{pmatrix}$$

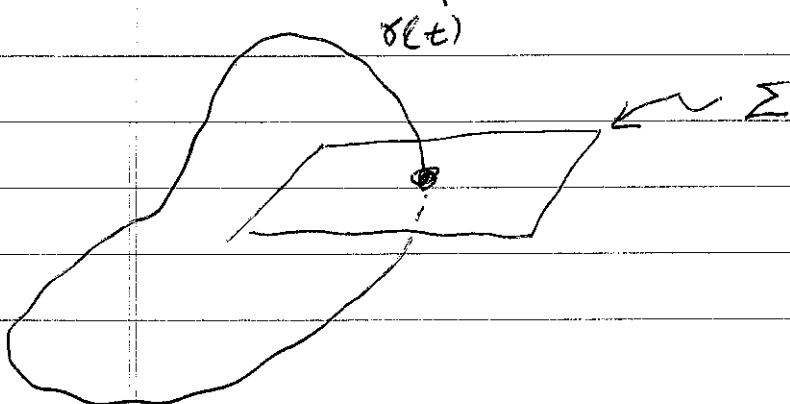
Some checks on M : $\mu_1 = 1$, $|\mu_2|, |\mu_3| < 1$

Another check: Abel's Identity:

$$\text{Det}(\Phi(t)) = \exp \left[\int_0^t \text{Tr}(A(s)) ds \right]$$

$$\Rightarrow \text{Det}(M) = \mu_2 \mu_3 = \exp \left[\int_0^T (a + x(s) - c) ds \right] > 0$$

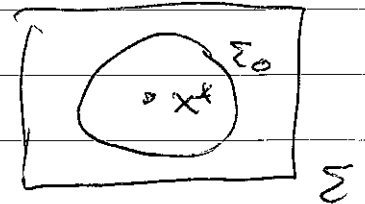
Alternative approach to linear stability: Poincaré return map



$$V(x) \cdot \hat{n} \Big|_{x \in \Sigma} > 0$$

$\hat{n}$ = surface normal to Σ

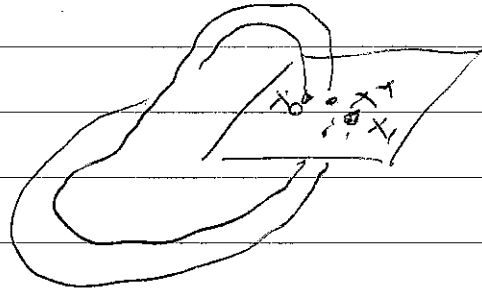
Poincaré return map P
 $P: \Sigma_0 \rightarrow \Sigma, \Sigma_0 \subset \Sigma$



$P(x^*) = x^*$ is a pt. on the limit cycle $\gamma(t)$ that is in Σ

$$P(x_0) = x_1$$

$$= \varphi_{T(x_0)}(x_0)$$



$T(x_0)$ = first return time to Σ

(expect $T(x_0) \approx T$ for x_0 close to x^*)

$P(x)$ is an $(n-1)$ -dimensional map.

$$x_{k+1} = P(x_k), \quad x^* = P(x^*)$$

linearize about $x = x^*$: $x_k = x^* + y_k$

$$x_{k+1} = x^* + y_{k+1}$$

$$\underline{x_{k+1}} = \underline{x^*} + y_{k+1} = P(x^* + y_k)$$

$$= \underline{P(x^*)} + DP(x^*) y_k + \dots$$

$$y_{k+1} = DP(x^*) y_k$$

is linearized map & its eigenvalues tell you about linear stability of $\gamma(t)$.

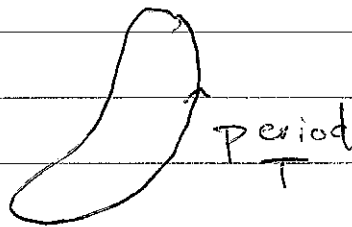
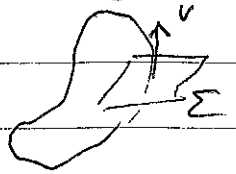
Thm 4.55 in text (p. 151)

$$\text{spec}(M) = \text{spec}(\text{DP}(x^*)) \cup \{1\}$$

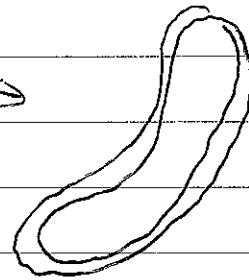
n Floquet multipliers
of Monodromy
matrix M

$(n-1)$ eigenvalues
of linearized
 $(n-1)$ -dim Poincaré
return map

associated
with
perturbations
along δ .

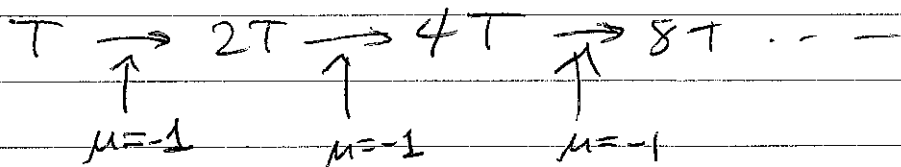


$\mu \rightarrow -1$



period $\approx 2T$

period-doubling cascade as you increase c .



$\mu = -1$ is associated with a "period-doubling"
bifurcation
(It requires $n \geq 3$.)

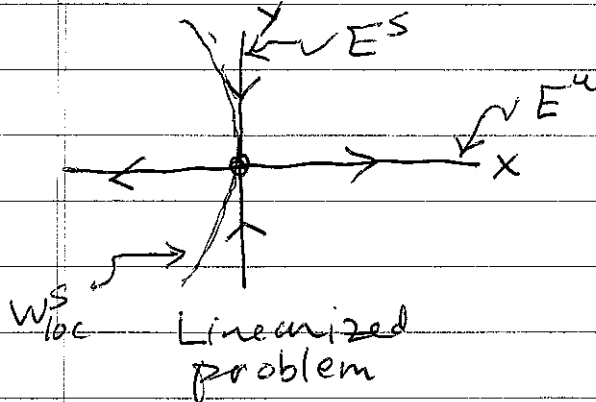
Approximating $W_{loc}^S(x^*)$ & $W_{loc}^u(x^*)$

Ex. $\left. \begin{aligned} \dot{x} &= 2x + y^2 \\ \dot{y} &= -2y + x + y^2 \end{aligned} \right\}$ EX. 5.12 in text

$x=y=0$ is fixed pt. of flow.

Linearize about it: $\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$

It is a saddle with 1-d. stable & 1-d. unstable manifolds.



$W_{loc}^S(0)$ is a graph over E^s

$$W_{loc}^S(0) = \{ (x, y) = (h_s(y), y), |y| < \delta \}$$

What do we know about $h_s(y)$ from stable manifold thm?

• It is tangent to E^s at $(0, 0)$

$$\Rightarrow h_s(0) = h'_s(0) = 0$$

It is smooth since f is smooth & it is invariant: if $(x_0, y_0) = (h_s(y_0), y_0)$, then $\varphi_t(x_0, y_0) = (h_s(y(t)), y(t))$ $t > 0$.

$$x(t) = h_s(y(t))$$

$$\Rightarrow \dot{x} = h'_s(y) \dot{y}$$

$$2x + y^2 \Big|_{x=h_s(y)}$$

$$-2y + x + y^2 \Big|_{x=h_s(y)}$$

$$\begin{cases} 2h_s + y^2 = h'_s (-2y + h_s + y^2) \\ h_s(0) = h'_s(0) = 0 \end{cases}$$

This is only guaranteed to be valid for y in some neighborhood of $y=0$, so can ~~not~~ approximate using power series?

$$h_s(y) = \cancel{\alpha} + \cancel{\beta} y + \gamma y^2 + \delta y^3 + \dots$$

$$2(\gamma y^2 + \delta y^3 + \dots) + y^2 = (2\gamma y + 3\delta y^2 + \dots)(-2y + y^2 + \gamma y^2 + \delta y^3 + \dots)$$

$$\mathcal{O}(y^2): \quad 2\gamma + 1 = -4\gamma$$

$$\Rightarrow \gamma = -\frac{1}{6}$$

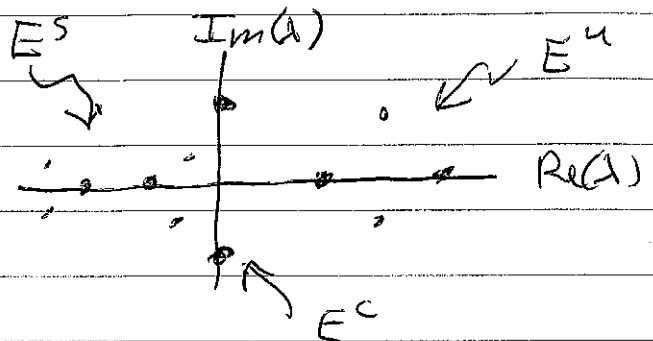
& so on

Non-hyperbolic fixed-pts & center manifolds

$$\dot{x} = V(x)$$

$$V(0) = 0$$

λ = eigenvalues of
 $D_x V(0)$



eigenspaces of $D_x V(0)$
 $E^s \oplus E^c \oplus E^u$

Center Manifold Thm: There is a

neighborhood of the origin where $\exists$
 C^k locally invariant manifolds:

W_{loc}^s , tangent to E^s , on which $|x(t)| \rightarrow 0$
 as $t \rightarrow \infty$; W_{loc}^u , tangent to E^u , on
 which $|x(t)| \rightarrow 0$ as $t \rightarrow -\infty$, and a
 local center manifold W_{loc}^c tangent to E^c

- W_{loc}^s & W_{loc}^u are unique, but W_{loc}^c need not be
- asymptotic behavior is unspecified for solns. in W_{loc}^c